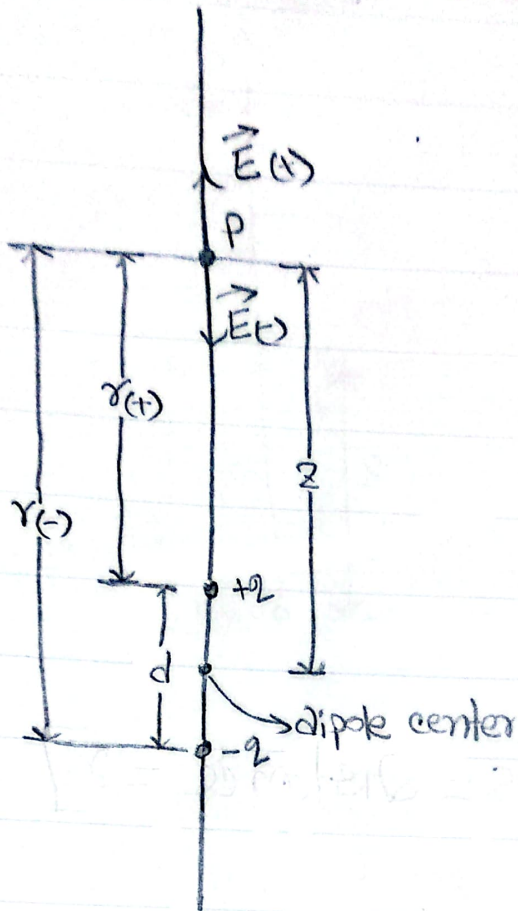


①

Date.....

Electric Field Due to an Electric Dipole.



Here,

$$r_+ = z - \frac{d}{2}$$

$$r_- = z + \frac{d}{2}$$

Electric field at point P,

$$E = E_+ - E_-$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r_+^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_-^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{(z - \frac{d}{2})^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(z + \frac{d}{2})^2}$$

$$\Rightarrow E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3}$$

$$\therefore E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$$

(2)

Potential Due to a Charge Particle
we know,

$$dW = \vec{F} \cdot d\vec{s}$$

$$\Rightarrow dW = q_0 \vec{E} \cdot d\vec{s}$$

$$\Rightarrow W = \int_i^f q_0 \vec{E} \cdot d\vec{s}$$

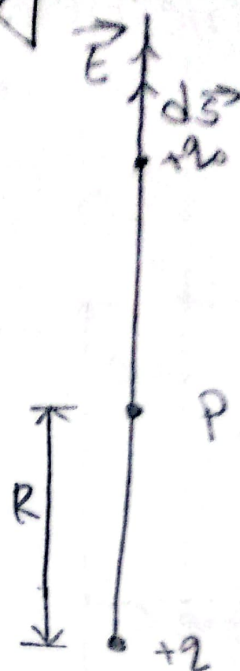
$$\Rightarrow \frac{W}{q_0} = \int_i^f \vec{E} \cdot d\vec{r} \quad [s = \text{distance} = r]$$

$$\Rightarrow V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{r} = - \int_i^f E \cdot dr \cos \theta$$

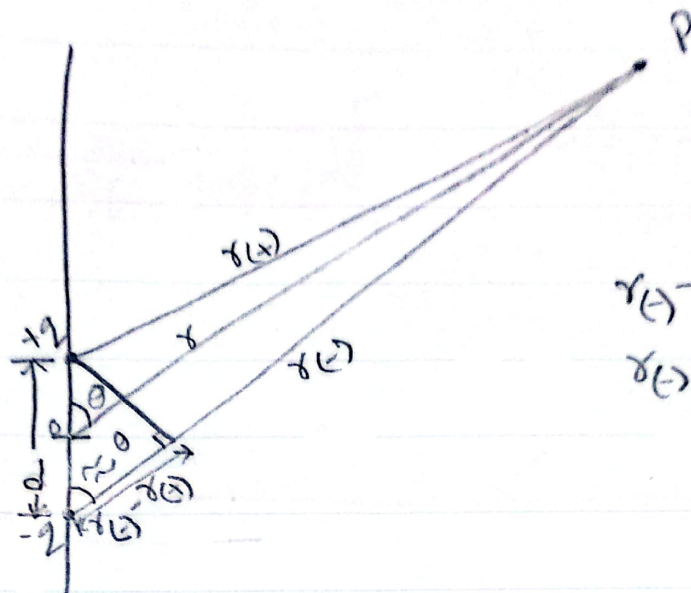
$$\Rightarrow 0 - V = - \int_R^\infty E \cdot dr \quad \therefore [\theta = 0^\circ]$$

$$\begin{aligned} \Rightarrow V &= \frac{q}{4\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} dr \\ &= \frac{-q}{4\pi\epsilon_0} \left. \frac{1}{r} \right|_R^\infty \end{aligned}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$



Potential Due to an Electric Dipole



$$r(-) - r(+)= d \cos \theta$$

$$r(-) \cdot r(+)= r^2$$

$$V = V_{(+)} + V_{(-)}$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r(+)} - \frac{q}{r(-)} \right)$$

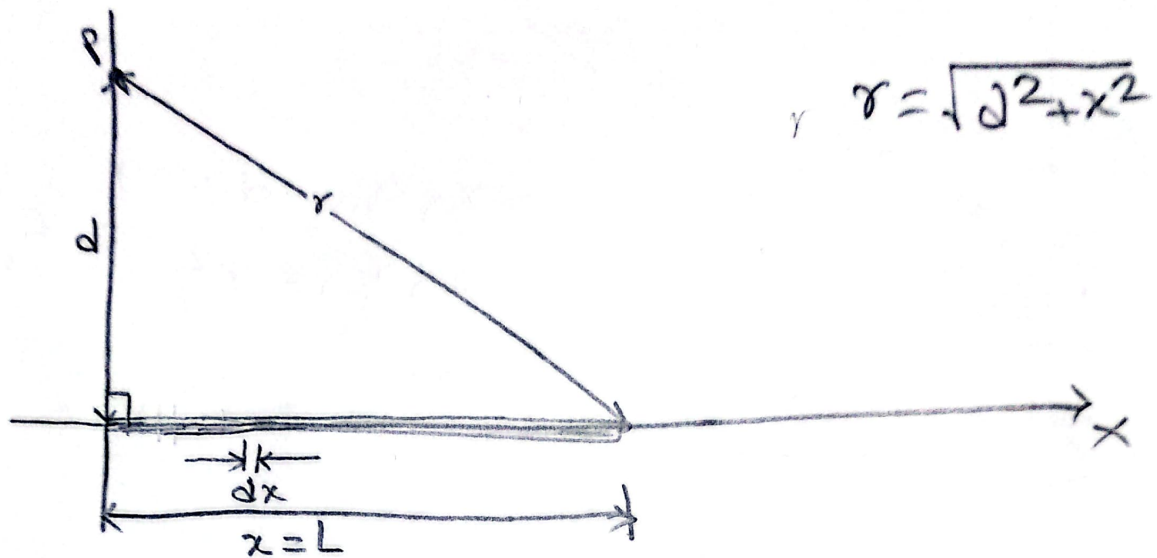
$$\Rightarrow V = \frac{q}{4\pi\epsilon_0} \frac{r(-) - r(+)}{r(-)r(+)}$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

(4)

Potential Due to A Continuous Charge Distribution



$\lambda = \text{charge density}$

$$\therefore dq = \lambda dx$$

$$\therefore dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x^2 + d^2)^{3/2}}$$

$$\therefore V = \int dV = \int_0^L \frac{1}{4\pi\epsilon_0} \frac{\lambda}{(x^2 + d^2)^{3/2}} dx$$

$$\Rightarrow V = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(x^2 + d^2)^{3/2}}$$

(5)

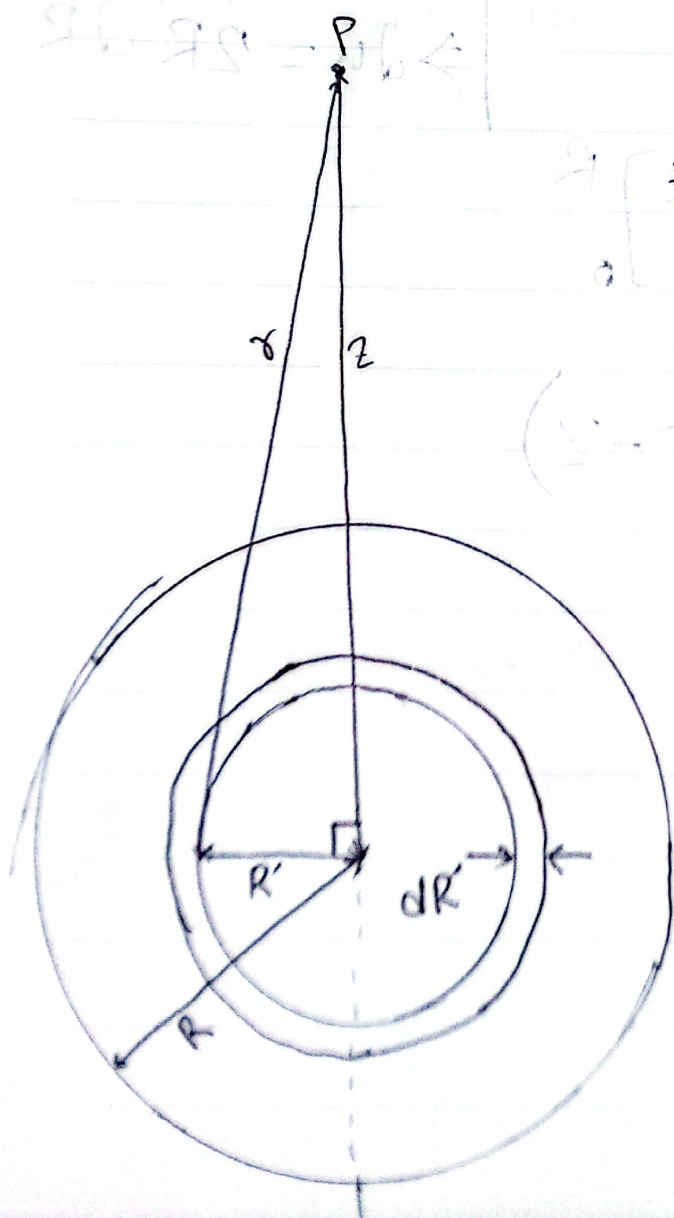
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$$\Rightarrow V = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + (x^2 + d^2)^{1/2}) \right]_0^L$$

$$\Rightarrow V = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(L + (L^2 + d^2)^{1/2}) - \ln d \right]$$

$$\therefore V = \frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{L + \sqrt{L^2 + d^2}}{d} \right)$$

Potential due to a Charge Disk.



$$q = \sigma A$$

$$\Rightarrow \frac{dq}{dr} = \sigma \frac{dA}{dr}$$

$$\Rightarrow \frac{dq}{dr} = \sigma \frac{d(\pi r^2)}{dr}$$

$$\Rightarrow \frac{dq}{dr} = \sigma 2\pi r$$

$$\therefore dq = \sigma 2\pi R' dR' [r=R']$$

$$\text{Here, } r = \sqrt{R'^2 + z^2}$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi R' dR'}{\sqrt{z^2 + R'^2}}$$

(6)

$$\therefore V = \int \phi dV = \int_0^R \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi R' dR'}{\sqrt{z^2 + R'^2}}$$

$$\Rightarrow V = \frac{\sigma \phi \pi}{4\pi\epsilon_0} \int_0^R \frac{2R'}{\sqrt{z^2 + R'^2}} dR'$$

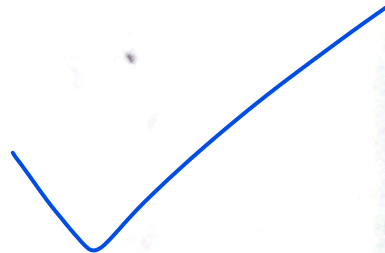
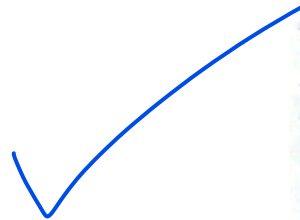
$$\Rightarrow V = \frac{\sigma}{4\epsilon_0} \int_0^R \frac{du}{\sqrt{u}}$$

$$= \frac{\sigma}{4\epsilon_0} \left[\frac{u^{1/2}}{1/2} \right]_0^R$$

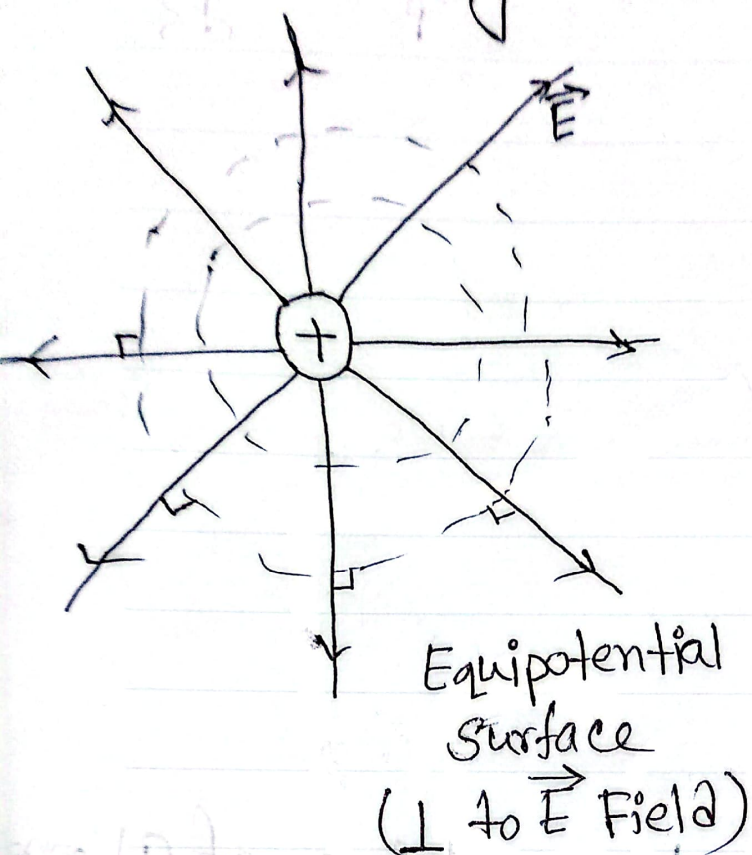
$$= \frac{\sigma}{2\epsilon_0} \left[\sqrt{z^2 + R'^2} \right]_0^R$$

$$\therefore V = \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z)$$

Let, $u = z^2 + R'^2$
 $\Rightarrow du = 2R' dR'$



Calculating Field from Potential.



$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$\Rightarrow \frac{dV}{dr} = \frac{d}{dr} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right)$$

$$= \frac{q}{4\pi\epsilon_0} \frac{d}{dr} (r^{-1})$$

$$= \frac{-q}{4\pi\epsilon_0} \frac{1}{r^2}$$

$$\therefore E = -\frac{dV}{dr}$$

Alternative

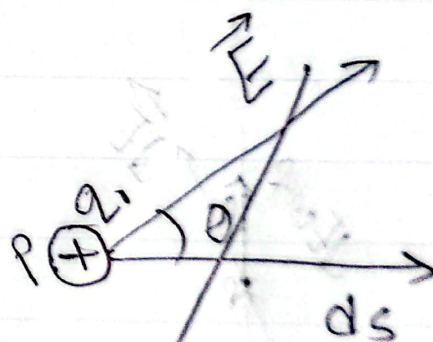
$$V = -\frac{w}{q}$$

$$\Rightarrow dW = -q dV$$

$$\Rightarrow -q dV = q_0 E \cos\theta ds$$

$$\Rightarrow E \cos\theta = -\frac{dV}{ds}$$

$$\therefore Es = -\frac{dV}{ds}$$



$$dw = -q_0 dV$$

$$\Rightarrow dw =$$

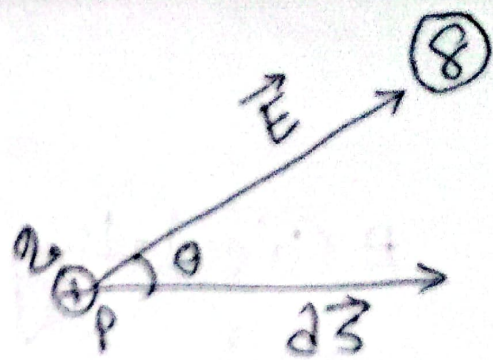
$$\Rightarrow \vec{F} \cdot d\vec{s} = -q_0 dV$$

$$\Rightarrow q_0 \vec{E} \cdot d\vec{s} = -q_0 dV$$

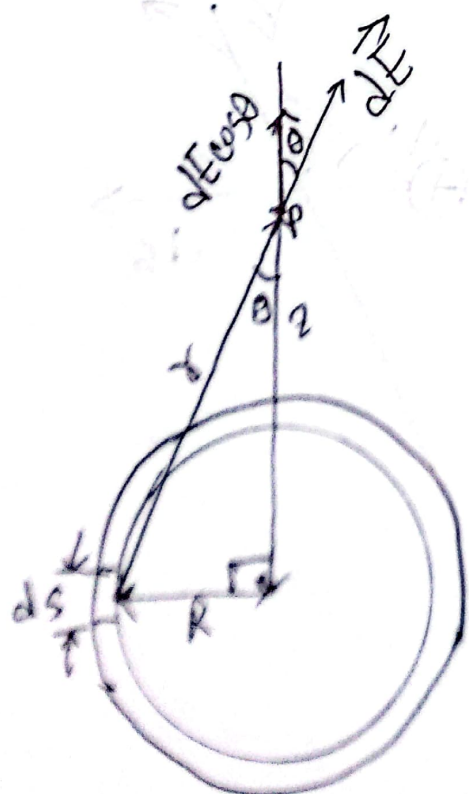
$$\Rightarrow q_0 E ds \cos \theta = -q_0 dV$$

$$\Rightarrow E \cos \theta = -\frac{dV}{ds}$$

$$\therefore E = -\frac{\delta V}{\delta s}$$



Electric Field Due to a Line of Charge



$$dq = \lambda ds$$

$\lambda \rightarrow$ Charge density

$$\cos \theta = \frac{z}{r}$$

$$= \frac{z}{\sqrt{z^2 + R^2}}$$

$$r^2 = z^2 + R^2$$

⑨

Date: 10/10/2017

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

$$\Rightarrow dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}$$

$$\Rightarrow dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)}$$

$$\Rightarrow dE \cos\theta = \frac{1}{4\pi\epsilon_0} \frac{2\lambda ds}{(z^2 + R^2)^{3/2}}$$

$$\Rightarrow \int dE$$

$$\therefore E = \int dE \cos\theta$$

$$= \int \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{(z^2 + R^2)^{3/2}} ds$$

$$= \frac{2\lambda}{4\pi\epsilon_0} \cdot \frac{2\lambda}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds$$

$$= \frac{2\lambda (2\pi R)}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

$$\therefore E = \frac{qz}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

$$[q = 2\pi R \lambda]$$

Electric Field Due to a Charge Disk (10)

$$q = \sigma A$$

$$\Rightarrow \frac{dq}{dr} = \sigma \frac{d(\pi r^2)}{dr}$$

$$\Rightarrow dq = \sigma (2\pi r dr)$$

$$dE = \frac{2\sigma \pi r dr}{4\pi \epsilon_0 (z^2 + r^2)^{3/2}}$$

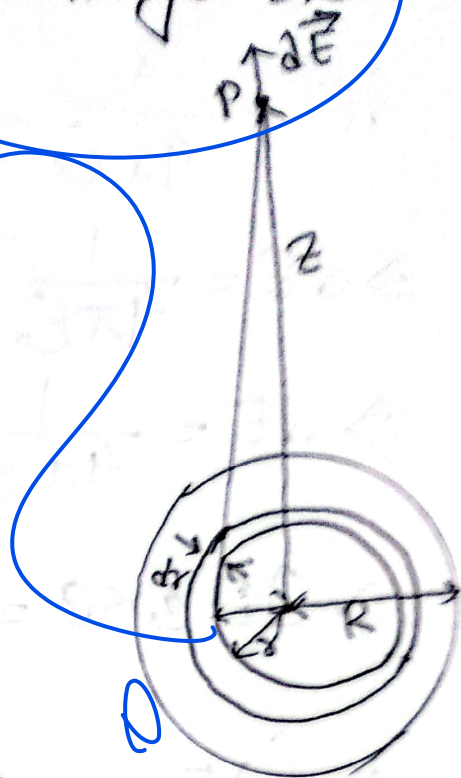
$$\Rightarrow E = \int dE = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + r^2)^{-3/2} (2r) dr$$

$$= \int_0^R u^{-3/2} du \times \frac{\sigma z}{4\epsilon_0}$$

$$= \frac{\sigma z}{4\epsilon_0} \left[\frac{u^{-1/2}}{-1/2} \right]_0^R$$

$$= -\frac{\sigma z}{2\epsilon_0} \left[(z^2 + R^2)^{-1/2} \right]_0^R$$

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$



$$\text{Let, } u = z^2 + r^2$$

$$\Rightarrow du = 2r dr$$